

The water-trial planning board

Teacher guide and worked answers

Plan a controlled comparison of water amount and plant growth, with repeated plants and consistent records.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsb/inputs/Build SOW 2026-2027.xlsx](#) | BUILD Weekly - Summer | C33

Plan an investigation into what plants need (links ASDAN allotment).

Chronological continuation of the supplied SOW. This lesson develops or reviews the named outcome; it does not claim an accreditation award.

Prior learning

Plants need suitable water.

Height and leaf condition provide different clues.

Success evidence

State a testable water-amount question.

Keep relevant conditions comparable.

Plan repeated plants, dates and a height rule.

Equipment

Printed shared evidence and one chosen pupil route; pencil; optional ruler; projector or offline HTML.

Optional six matched, established young bean seedlings in labelled 200 mL pots with drainage; trays; ruler; measuring syringe or cylinder; adult-approved compost.

Preparation

Print the shared evidence, one response route and exit page. Routes are alternatives, not a workload ladder.

Read the worked answers. Prepare an oral, pointing or adult-scribed route where useful.

The supplied fictional plan compares 5 and 15 mL daily for 10 days. These are teaching values, not universal watering recommendations. For a real investigation, an adult pilots suitable amounts for the species, pots and room, records any changes and checks plant welfare daily.

Optional setup can begin today: label plants A1-A3 and B1-B3, record baselines, allocate comparable seedlings across groups, and agree who maintains the record. Future observation time must be scheduled separately.

Practical care

Use non-toxic plants; no tasting or direct handling of fertiliser. Wash hands after handling plants or compost.

Keep water in trays away from electrical devices; wipe spills. Use needle-free measuring equipment.

An adult checks plants daily. Stop or adjust a real treatment if severe stress develops; record the change honestly. Do not overheat or seal plants in a cupboard.

Ready to teach alternative

All core reasoning uses the supplied evidence and can be completed in 40 minutes without a live practical.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A good plan can be followed by someone else
2	Retrieval	2-5	Retrieve what makes a useful comparison
3	I do	5-12	I repair two changes
4	I do	Within stage	I define the measurement before taking it
5	I do	Within stage	I make dates and responsibilities visible
6	We do	12-16	The plan-repair meeting
7	We do	16-20	Prediction or result?
8	Hinge	20-23	Which plan best tests water amount?
9	You do	23-25	Choose your science route
10	You do	25-35	Make your own evidence record
11	You do	Within stage	Check what the evidence can show
12	Review	35-38	Lundy loop: improve the shared account
13	Review	Within stage	Keep the useful change
14	Exit	38-40	Three final checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A good plan can be followed by someone else

Explain the difference between planning now and waiting for growth later. All pupils can meet today's planning objective on paper.

2 Retrieval Retrieve what makes a useful comparison

1 valid requirement. 2 no; leaf colour/condition also useful. 3 centimetres.

3 I do I repair two changes

Think aloud: "If the groups differ, I would not know whether water or darkness mattered. I repair the light conditions before I start." Other factors still need matching; use three comparable seedlings in each group.

4 I do I define the measurement before taking it

Shares I do time. Think aloud: "Including the pot would add something I am not studying. I choose one start and endpoint and use them every time." Worked reading is 6 cm.

5 I do I make dates and responsibilities visible

Shares I do time. Think aloud: "A results heading is not a result. I leave future values blank and plan who will measure." Keep the same time of day; rotate positions within a comparable light area where useful and record method.

6 We do The plan-repair meeting

W1: daily water amount; height change and leaf condition; two relevant controls. Have one pupil challenge the plan and another point to a fix in the shared cards.

7 We do Prediction or result?

W2: expectation is a prediction; the measured statement is only a result if actually measured. The class has not produced those data today. Choose a useful blank record heading and a baseline task.

Reveal: Predict in a separate space. Record results only after observation.

8 Hinge Which plan best tests water amount?

Correct first option; all options need a reason.

A. Change water amount; keep species and growing conditions comparable. - This isolates the planned water comparison more effectively.

B. Change water amount, species and light together. - Several factors could explain the difference.

C. Write the expected heights into the results table now. - Predictions are not measurements.

9 You do Choose your science route

Set ONE route. Planning is today's assessed product. An optional class setup must use adult-piloted care levels; paper plans remain fully valid. Do not let a pupil fill unknown results to make the plan look finished.

10 You do Make your own evidence record

Planning is today's assessed product. An optional class setup must use adult-piloted care levels; paper plans remain fully valid. Do not let a pupil fill unknown results to make the plan look finished.

11 You do Check what the evidence can show

Shares independent time. Ask for a reason, not just a correct word. Planning is today's assessed product. An optional class setup must use adult-piloted care levels; paper plans remain fully valid. Do not let a pupil fill unknown results to make the plan look finished.

12 Review Lundy loop: improve the shared account

Listen to a selected pupil revision and visibly adopt a scientifically accurate contribution. Offer private, written or partner feedback.

Reveal: Keep one planned change, matched groups, repeated plants and a consistent dated record.

13 Review Keep the useful change

Shares review time. Name how a pupil contribution changed a label, method or conclusion. Do not rank pupils.

Reveal: Keep one planned change, matched groups, repeated plants and a consistent dated record.

14 Exit Three final checks

Collect brief individual evidence. Use the answer key and re-teach the specified misconception if needed.

Answers and assessment

Retrieval 1-3

1 valid requirement. 2 no: leaf condition/colour also matter. 3 cm.

We do W1-W2

W1 water amount; height change plus leaf condition; relevant matched conditions including species, starting size, pot/compost, light or temperature. W2 prediction is expected; a result comes from an observation or measurement actually made.

Hinge

First option correct. Second has confounds; third invents evidence.

S1-S5

S1 water amount only. S2 two valid controls. S3 three. S4 soil surface to main shoot tip. S5 observed later.

M1-M4

M1 how does daily water amount affect these young bean seedlings over 10 days? M2 5 versus 15 mL daily in the fictional plan; two valid controls. M3 label A1-A3/B1-B3; match and allocate seedlings; record baseline; apply planned water; measure soil-to-tip in cm and leaf condition on days 0/3/7/10 at consistent times; keep all records. M4 any reasoned prediction clearly labelled, plus daily care check and adult action if stress.

T1-T4

T1 species, starting size and light all differ; water is intended change. T2 three comparable seedlings of one species per group, matched pots/compost and growing conditions, different water amounts only. T3 dated baseline and agreed repeat observations with consistent soil-to-shoot-tip cm measurement and leaf notes. T4 prioritise suitable care, stop/adjust treatment as necessary, record exactly what changed; interpret the result with this limitation.

Exit E1-E3

E1 daily water amount. E2 individuals vary; repeats make comparison less dependent on one unusual plant. E3 no.

R1-R2

Responses vary. Credit a scientific revision, evidence-based defence or relevant next question. A private or oral response has equal value.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, independent 12, review 3, exit 2. Zero-minute slides share the named stage time.

BUILD uses the supplied Year 3 science anchor with age-respectful contexts for older pupils. Supported, standard and stretch are selectable routes within the pathway.

Interactive We do: predict first, test against the controlled SVG model or supplied evidence, then explain or revise. Shared W tasks offer the equivalent paper discussion. Animation is a teaching representation, not filmed experimental evidence.

Keep constructed evidence separate from actual class observations. Do not invent measurements or require internet research.

A real 10-day study cannot be completed within this 40-minute lesson. The next lesson supplies dated constructed data for complete analysis without pretending that time has passed. Optional real observations continue separately.

Secure core: an explicit question, one intended change, relevant controls, repeat plants and honest records.

Access and responsive teaching

Supported

Short choices, word bank and pointing or adult-scribed reasons; same core scientific goal.

Standard

Make a short evidence-based explanation or record using the supplied information.

Stretch

Apply the core idea to a changed case, evaluate evidence or identify a limit.

Regulation

Preview the sequence. Offer seated observation, quiet paired rehearsal, private feedback and a pass from public sharing. Routes respond to current access needs, not a diagnosis.

Misconceptions to check

Changing water and light tests water alone.

Two changes prevent separating their effects.

Check: Ask what else could explain a difference.

A prediction belongs in the results column.

Predictions are expectations; results are actual measurements.

Check: Ask whether the measurement has happened.

One plant represents all plants.

Individuals vary; matched repeats improve a comparison.

Check: Ask why each group has three seedlings.

Word	Meaning
change	The factor deliberately made different.
measure	Find and record a quantity.
control	A condition kept comparable.
repeat	Use further observations or plants to check variation.

Scientific references

[Department for Education: science programmes of study](#)

Checked 8 September 2026. Year 3 plants, rocks, light and forces underpin the SOW. Accessible teaching models are adapted to the named lesson objective.

Royal Horticultural Society: sowing seeds indoors

Checked 8 September 2026. Germination uses stored reserves and requires appropriate water, oxygen and temperature; seedling care differs from starting germination.

Science and Plants for Schools: plant biology animations

Checked 8 September 2026. Scientific background for our original simplified diagrams: photosynthesis, growth and transport. No external animation is required or embedded.